TELECOM CUSTOMER CHURN ANALYSIS

Python • MySQL • Power BI

Project Report

Domain	Telecommunications
Dataset	Telco Customer Churn
Dataset size	7,043 customer records and 21 original columns; one derived tenure-group column added during preparation.
Tools	Python (Pandas, NumPy), MySQL, Power BI

Prepared as an end-to-end data analytics capstone project.

1. Executive Summary

Customer churn occurs when customers stop using a company's services. Understanding churn helps a telecommunications provider identify customer groups that may need better service, clearer pricing, or retention offers. This project uses a customer-level telecom dataset to prepare, analyse, query, and visualise churn information through a Python–MySQL–Power BI workflow.

The dataset contains 7,043 customer records. The analysis identified 1,869 customers marked as churned and 5,174 marked as retained. This corresponds to a churn rate of approximately 26.54%. The average tenure for churned customers is about 17.98 months, compared with 37.57 months for retained customers. These results suggest that churn is associated with shorter customer relationships in this dataset.

The work is organised into three main stages: Python data preparation and exploratory analysis, SQL-based business analysis and reusable views, and a Power BI report with KPI cards, slicers, and charts. The dashboard is designed to help a viewer compare churn patterns across contract types, tenure groups, payment methods, internet service, and other customer attributes.

Project objectives

Objective	How it is addressed
Prepare reliable data	Inspect missing values, duplicates, data types, blank TotalCharges values, and derived tenure groups in Python.
Answer business questions	Use MySQL aggregations and views to summarise churn across customer segments.
Communicate findings	Use Power BI KPI cards, slicers, and comparison charts in a two-page dashboard.
Support decisions	Highlight customer groups and service patterns for further retention investigation.

> *Scope note: this is a descriptive analytics project. The results describe patterns in the supplied dataset and do not, by themselves, prove why a customer churned.*

2. Dataset and Business Context

The project uses a public Telco Customer Churn dataset commonly used for customer-retention analysis. Each row represents one customer. The source data includes customer identifiers, demographic indicators, subscribed services, contract and payment information, account tenure, monthly charges, total charges, and the churn outcome.

Dataset overview

Item	Description
Rows	7,043 customer records
Original columns	21 columns in the raw dataset
Target field	Churn, with Yes/No values indicating churn status
Customer/account fields	customerID, gender, SeniorCitizen, Partner, Dependents, tenure
Service fields	PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies
Billing fields	Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges
Derived field	TenureGroup, created to group tenure into readable ranges for analysis and visualisation

Business problem

A telecom provider needs to understand how churn differs by contract, payment method, service subscription, tenure, and customer characteristics. A summary churn percentage alone is not enough: segmentation can reveal which groups deserve closer investigation. The project therefore combines descriptive statistics with grouped SQL queries and interactive dashboard filters.

Questions explored

- How many customers are in the dataset, and how many churned or remained?
- How does churn vary by contract type and internet service?
- How do average monthly charges and tenure differ by churn status?
- Which payment methods and customer segments show different churn counts or rates?
- How can these comparisons be explored interactively in a dashboard?

3. Methodology and Workflow

The project follows a practical analytics pipeline. Python is used to inspect and prepare the raw CSV, MySQL stores the cleaned customer table and provides reusable query results, and Power BI presents the findings visually.

Stage	Activities	Output
1. Load	Read the raw CSV using Pandas and inspect its dimensions, columns, and sample rows.	Loaded DataFrame
2. Clean	Check missing values and duplicates; examine data types; handle blank TotalCharges values; convert the field to numeric.	Cleaned dataset
3. Analyse	Calculate customer and churn totals, grouped churn counts, average tenure, charge summaries, and outlier checks.	Python analysis results
4. Store	Import the cleaned CSV into the customer_churn_analysis MySQL database.	cleaned_customer_churn table
5. Query	Run business questions and create views for churn summaries and segment comparisons.	SQL results and 10 views
6. Visualise	Connect Power BI to the cleaned table, create DAX measures, and build two report pages.	Interactive dashboard

Tools used

- Python: Pandas for data manipulation and NumPy for numerical checks.
- MySQL Workbench: database queries, grouped analysis, and views.
- Power BI Desktop: data visualisation, slicers, KPI cards, and DAX measures.

Data flow

> ***Raw CSV → Python cleaning and analysis → Cleaned CSV → MySQL table and views → Power BI model and dashboard***

The project is descriptive rather than predictive. No machine-learning churn model is claimed in this report.

4. Python Data Preparation and Analysis

Python was used to inspect the raw data, assess data quality, prepare fields for analysis, and calculate initial business summaries. The notebook is the reproducible record of the preparation workflow.

Data quality checks

Check	Observed result / action
Dataset dimensions	7,043 rows × 21 columns in the raw dataset
Missing values	The initial column-wise missing-value check returned zero for the columns as loaded; blank TotalCharges entries were also checked separately.
Duplicate records	Duplicate check returned 0.
TotalCharges	Blank strings were treated as missing, converted to numeric, and filled with 0 for records where no total charge was recorded; this is consistent with zero tenure in the dataset.
Data types	Data types were inspected and TotalCharges was converted to numeric for calculations.
Derived grouping	TenureGroup was created from tenure ranges for grouped analysis and dashboard visuals.
Outlier review	IQR checks were performed for MonthlyCharges and TotalCharges. Outliers were reviewed rather than automatically removed.

Derived tenure groups

The TenureGroup field groups customer tenure into four ranges: 0–12 months, 13–24 months, 25–48 months, and 49–72 months. These groups make it easier to compare customer churn across relationship duration. Group labels should be interpreted as descriptive categories, not as evidence of causation.

Initial analysis results

Metric	Result
Total customers	7,043
Churned customers	1,869
Retained customers	5,174
Churn rate	Approximately 26.54%
Average tenure — retained customers	37.57 months
Average tenure — churned customers	17.98 months

The notebook also examines churn by contract type and internet service, average monthly charges by churn status, and descriptive statistics for numeric fields. Charts are intentionally not required in the Python notebook because visual reporting is handled in Power BI.

5. MySQL Business Analysis

The cleaned data was imported into the customer_churn_analysis database in MySQL Workbench as cleaned_customer_churn. SQL queries were used to answer business questions with grouping, filtering, and summary calculations. Reusable views were created so that common summaries can be queried again without rewriting each calculation.

Business questions covered

#	Business question	Analysis approach
1	How many customers are in the dataset?	COUNT / total row count
2	How many customers churned versus stayed?	GROUP BY Churn
3	How does churn count vary by contract?	GROUP BY Contract and Churn
4	Which contract type has the highest churn count?	Grouped counts and ordering
5	How do average monthly charges differ by churn status?	AVG with grouping
6	How does churn vary by internet service?	GROUP BY InternetService and Churn
7	What is average tenure by churn status?	AVG(tenure) with grouping
8	How does churn differ by payment method?	GROUP BY PaymentMethod and Churn
9	How does churn differ by senior-citizen status?	GROUP BY SeniorCitizen and Churn
10	How do monthly charges vary by contract type?	AVG(MonthlyCharges) by Contract
11	How do contract churn counts compare with total customers and churn percentage?	Join contract summary results
12	Which contract groups rank highest by churn percentage?	RANK() window function

Reusable views

Ten views were created for summaries including overall churn, contract churn, internet-service churn, payment-method churn, tenure churn, senior-citizen churn, monthly charges by churn, and churn-rate summaries by payment method, contract, and tenure group.

SQL implementation notes

The project database contains the cleaned customer table and the views listed above. The query script should be saved as customer_churn_analysis.sql. When presenting results, confirm that the SQL file contains the final version of the queries and that the views can be listed in MySQL Workbench. Avoid re-running CREATE VIEW statements without replacing or dropping existing views if the script does not use CREATE OR REPLACE VIEW.

6. Power BI Dashboard and Measures

Power BI was used to build a two-page interactive report. The cleaned_customer_churn table is the primary source table. Since the current report uses one imported table, no relationship between separate model tables is required for the present design.

DAX measures

Measure	Purpose
TotalCustomers	Distinct count of customerID.
ChurnedCustomers	Distinct count of customers where Churn = Yes.
RetainedCustomers	Distinct count of customers where Churn = No.
ChurnRate	Churned customer count divided by total customer rows, formatted as a percentage.
AverageMonthlyCharges	Average of MonthlyCharges.
AverageTenure	Average of tenure in months.

Page 1 — Customer Churn Overview

The overview page contains KPI cards for total customers, churned customers, retained customers, churn rate, and average monthly charges. Slicers allow filtering by Contract, PaymentMethod, and Churn. Charts compare customer churn by contract type, tenure group, and payment method.

Page 2 — Detailed Churn Analysis

The detailed page provides additional comparisons, including churn by internet service, senior-citizen status, tech support, and average monthly charges by contract type. The slicers are copied to the second page to keep the interaction style familiar. Slicer selections were tested to confirm that the charts respond to the filters.

Design choices

The report uses a consistent, restrained visual style: aligned visuals, matching sizes where practical, light borders, white backgrounds, and a consistent blue colour palette. Borders separate the visual containers without adding heavy outlines around individual bars. The aim is to keep the dashboard readable and focused on the data.

Suggested dashboard screenshot placement

> *Insert a screenshot of the final Overview page and the Detailed Analysis page here before submission if screenshots are required by the course. The report text describes the completed dashboard but does not embed screenshots because none were supplied for this report file.*

7. Key Findings and Recommendations

Key findings

Finding	Interpretation
Overall churn is approximately 26.54%.	Around one in four customers in the dataset is marked as churned, making churn a material business issue to investigate.
1,869 customers churned; 5,174 were retained.	The retained group is larger, but the churned group is substantial enough to merit segment-level analysis.
Average tenure is lower among churned customers.	Churned customers average about 17.98 months of tenure versus 37.57 months for retained customers. Shorter tenure is associated with churn in this dataset.
Contract, payment, service, and tenure segments can be compared.	The SQL views and Power BI slicers support exploration of where churn counts or rates differ between customer groups.

Recommendations for a telecom business

1. Investigate early-tenure churn: review onboarding, service setup, customer support, and early-life experience for customers with shorter tenure.
2. Compare contract segments using churn rates, not counts alone, because segment sizes may differ.
3. Review payment-method and service-level patterns alongside customer feedback before deciding on interventions.
4. Test retention offers with measurable success criteria, then compare churn rates before and after the intervention.
5. Refresh the analysis periodically with new customer data to see whether patterns remain stable.

Limitations

This project is based on one supplied dataset and describes associations rather than causal effects. The analysis does not include customer interviews, service-quality logs, campaign outcomes, or a predictive model. Outliers were reviewed but not automatically removed. Any business action should be validated with additional data and controlled measurement.

8. Conclusion and Deliverables

The Telecom Customer Churn Analysis project demonstrates an end-to-end analytics workflow across Python, MySQL, and Power BI. Python was used to inspect and prepare the dataset, MySQL was used to answer business questions and create reusable views, and Power BI was used to present the findings in an interactive two-page dashboard.

The main descriptive results are 7,043 customers, 1,869 churned customers, 5,174 retained customers, and an overall churn rate of approximately 26.54%. Average tenure is lower for churned customers than for retained customers. These findings provide a starting point for more focused retention analysis.

Submission checklist

Deliverable	Expected file / verification
Python notebook	customer_churn_analysis.ipynb — confirm notebook cells are saved and outputs are current.
SQL script	customer_churn_analysis.sql — include final business questions and view definitions.
Power BI report	customer_churn_analysis.pbix — save after final formatting and verify both report pages.
Project report	Customer_Churn_Analysis_Project_Report.pdf — this document.
Presentation	Prepare an 8–12 slide presentation summarising the same workflow and findings.
Evidence / source	Add final dashboard screenshots and dataset/source information if required by the instructor.

Final conclusion

The combined use of Python, MySQL, and Power BI makes the churn analysis traceable from raw data preparation to business-facing visualisation. The dashboard supports exploratory decision-making and provides a foundation for future work such as churn prediction, customer feedback analysis, and retention campaign evaluation.

End of Report